

Adaptive wavelet decomposition and event-aware high-frequency modeling network for multivariate time series forecasting

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ABSTRACT

Multivariate time series (MTS) forecasting has wide applications in real-world domains such as traffic, weather, and ocean monitoring. However, real-world MTS data often exhibit multiscale non-stationary patterns. These patterns arise from the dynamic coupling between long-term trends and sudden local events, making accurate forecasting highly challenging. Existing methods primarily rely on global modeling or fixed decomposition strategies. However, such approaches fail to adapt to varying spatiotemporal data and diverse task contexts. They cannot effectively disentangle trend and event sequences in accordance with the characteristics of time series data. Moreover, they cannot model unstable high-frequency events. To address these issues, we propose an Adaptive Wavelet Decomposition and Event-aware High-frequency Modeling Network (AweHF). The model employs an Adaptive Wavelet Decomposition module (AWD) to decouple the original sequence into low-frequency trends and high-frequency events in a data-driven way, avoiding the limitations of fixed wavelet bases. Subsequently, we apply a lightweight multilayer perceptron (MLP) to capture long-term dependencies in the trend component. In addition, we design a Time Aggregation Network (TAN) and Dual-Source Personalized Graph Convolution (DSPGC) to jointly model the volatility and instability of the event component. Finally, the bidirectional interaction fusion mechanism is used to integrate the trend and event components to fully exploit their complementary advantages. We conducted extensive experiments on six real-world datasets from multiple domains, and our results demonstrate that AweHF consistently outperforms all state-of-the-art baselines, achieving an average MAE reduction of more than 3.8% across the datasets. Code is available at this repository: <https://github.com/WangChengci/AweHF>.

1. Introduction

Multivariate time series (MTS) forecasting aims to identify patterns and correlations among variables within historical data, to predict future trends. MTS is extensively applied in fields such as traffic control [1,2], energy management [3,4], social services [5] and weather forecasting [6,7]. With the rapid development of emerging intelligent sensors, the quantity and complexity of time series data continue to grow. Accurate MTS forecasting optimizes resource allocation and greatly enhances the efficiency of social services and public safety.

In real-world scenarios, the MTS data do not follow a single smooth pattern. Instead, they exhibit multiscale non-stationary characteristics [8]. These characteristics originate from the dynamic coupling of low-frequency trends and high-frequency events. For example, as shown in Fig. 1(a), the traffic flow at the same node over time can reflect both regular commuting trends and sudden changes. Similarly, in oceanography,

as illustrated in Fig. 1(b), wave heights are affected by long-term seasonal trends and short-term variations caused by extreme weather [9,10]. Beyond the MTS domain, modeling complex systems with non-stationarity [11], multi-scale dynamics, and intricate interactions has long been a central challenge in machine learning. Paradigms such as neural networks [12,13], Gaussian process regression [14,15], graph-based learning [16–18], and ensemble methods have achieved notable success by leveraging adaptive representations, hierarchical decomposition, and hybrid modeling strategies [19,20]. These advances suggest a shared principle: heterogeneous and non-stationary processes are better captured by decomposing distinct patterns and modeling them with specialized, interacting components. Inspired by this perspective, we argue that MTS forecasting can similarly benefit from adaptive decomposition and heterogeneous modeling, particularly when facing coupled trends, abrupt events, and dynamic inter-variable dependencies. The mixing of trends and events makes it hard for a single model to fully capture all

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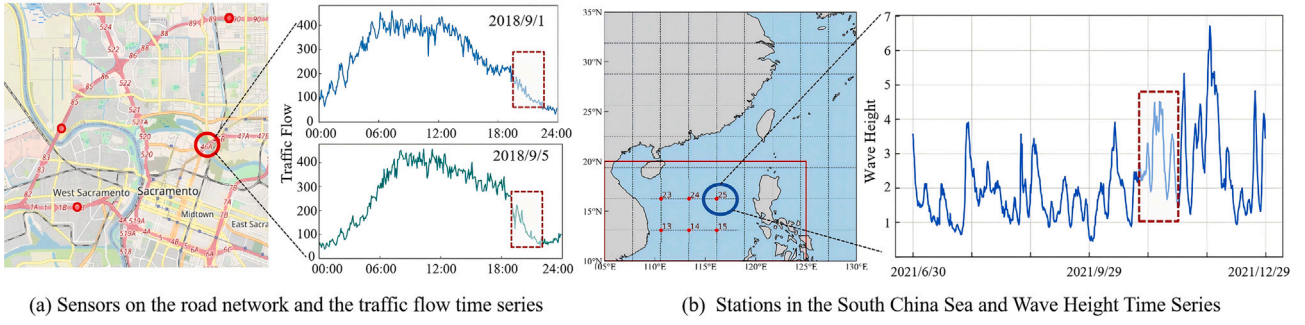


Fig. 1. Examples of non-stationary multivariate time series: (a) Traffic flow over a day; (b) Wave-height variation across a six-month period.

the features of MTS data. Therefore, it is necessary to separate trends and events based on their intrinsic data features and then study them individually.

Existing methods still face considerable limitations in handling these complex patterns. First, most methods adopt a global modeling strategy, ignoring the multiscale characteristics of data. For example, models such as STFGNN [21] and DDGCRN [22] directly take raw time series as input, but capture only superficial features. Although ASTGNN [23] enhances feature representation by encoding spatiotemporal embeddings, it overlooks sudden events that occur within long-term stable trends. Consequently, these models struggle to capture the interactions between long-term trends and short-term events. Inadequate upstream data processing severely limits downstream models from extracting spatiotemporal features, making accurate prediction difficult. This challenge has motivated researchers to extract richer information from raw data while designing complex downstream models to capture diverse characteristics. Some methods attempt to decompose time series to obtain more information; however, they have considerable limitations. For instance, DSTIGNN [16] uses downsampling to extract multi-resolution data, and D²STGNN [24] decouples the associations between nodes from the historical changes of individual nodes. However, these methods neglect the interplay between trends and events in time-series data and remain confined to superficial spatiotemporal decompositions [25]. Methods such as FEDformer [26] and Pathformer [27] decompose time series into trend and seasonal components. However, their coarse decoupling excessively smooths the fluctuations of sudden events, resulting in mediocre predictions because models find it difficult to capture local bursts. These methods underestimate the importance of two fundamental properties of time series: trends and events. Trends reflect long-term changes and intuitively indicate growth or decline over a period. Events represent sudden changes at specific moments, revealing the instability of the time series. HiLIN [28] and STWave [29] decompose time series using discrete wavelet transforms (DWTs) with fixed wavelet bases. However, fixed bases and windows cannot adapt to different datasets or varying periods within the same dataset, limiting their ability to represent abrupt, highly volatile, and non-periodic sequences. Moreover, high-frequency sequences are treated as ordinary sequences without modeling the abruptness and instability of high-frequency events. These abrupt components largely determine the difficulty of prediction, emphasizing the need for specialized methods that can capture high-frequency features and handle instability.

Given the limitations of existing methods in handling MTS data, we propose a novel **Adaptive wavelet decomposition and event-aware high-frequency modeling network (AweHF)**, as shown in Fig. 2. The proposed model applies adaptive wavelet decomposition module (AWD) to decompose coupled raw time series into low-frequency trend sequences and high-frequency event sequences. It dynamically adjusts the parameters of the wavelet transform to avoid the limitations of predefined wavelet bases and fixed windows, which lack adaptability to the specific task. This process provides downstream forecasting models with

low- and high-frequency sequences that are more task-relevant. We capture the unique properties of these sequences by employing a multilayer perceptron (MLP) to learn the long-term stable patterns of low-frequency trends and design a TAN-DSPGC module to model the instability of high-frequency events. Specifically, TAN-DSPGC consists of a Temporal Aggregation Network (TAN) that compresses the temporal dimension, providing global mutation awareness for the subsequent Dual-source Personalized Graph Convolution (DSPGC). Our study demonstrates that decomposition-based learning and personalized modeling of different sequence characteristics are crucial for MTS forecasting. In addition, the introduction of the bidirectional interaction fusion framework and dynamic graphs enables our proposed model to recognize diverse patterns across different data types. The main contributions of this study can be summarized as follows:

- **AWD for trends and events.** We introduce AWD to decompose MTS into low-frequency trends and high-frequency events. By dynamically adjusting wavelet parameters, the decomposition adapts to heterogeneous data characteristics, enabling accurate modeling of both stable trends and abrupt events.
- **TAN-DSPGC for high-frequency events.** We design the TAN-DSPGC module to capture the abruptness and instability of high-frequency sequences. A multiscale TAN compresses the temporal dimension to provide global mutation awareness. An attention mechanism adaptively highlights nodes with significant changes, while DSPGC models latent inter-node influences, enabling precise high-frequency event modeling.
- **Comprehensive empirical validation.** AweHF conducts extensive prediction experiments on six real-world datasets. The results confirm the superior performance of AweHF compared with state-of-the-art baselines. In addition, ablation studies are conducted to verify the contribution and effectiveness of each module.

2. Related work

2.1. Multivariate time series prediction

Early research on MTS forecasting primarily relied on statistical methods. AutoRegressive Integrated Moving Average (ARIMA) [30] which combines the AutoRegressive (AR) and Moving Average (MA) components provides flexible and reliable statistical capabilities. Similarly, Vector AR (VAR) [31] generalizes the AR model, and is suited for MTS forecasting tasks. However, such methods assume linear or simple nonlinear dependencies, which limit their ability to model complex interactions among variables. As recurrent neural network (RNN) variants, long short-term memory (LSTM) [32] and gated recurrent unit (GRU) [33] excel at capturing long-term dependencies. Nevertheless, their sequential nature hinders parallelization and poses challenges related to the vanishing gradient problem. Spatiotemporal Graph Neural Networks (STGNNs) integrate graph convolution with temporal modules to learn spatiotemporal dependencies among multiple

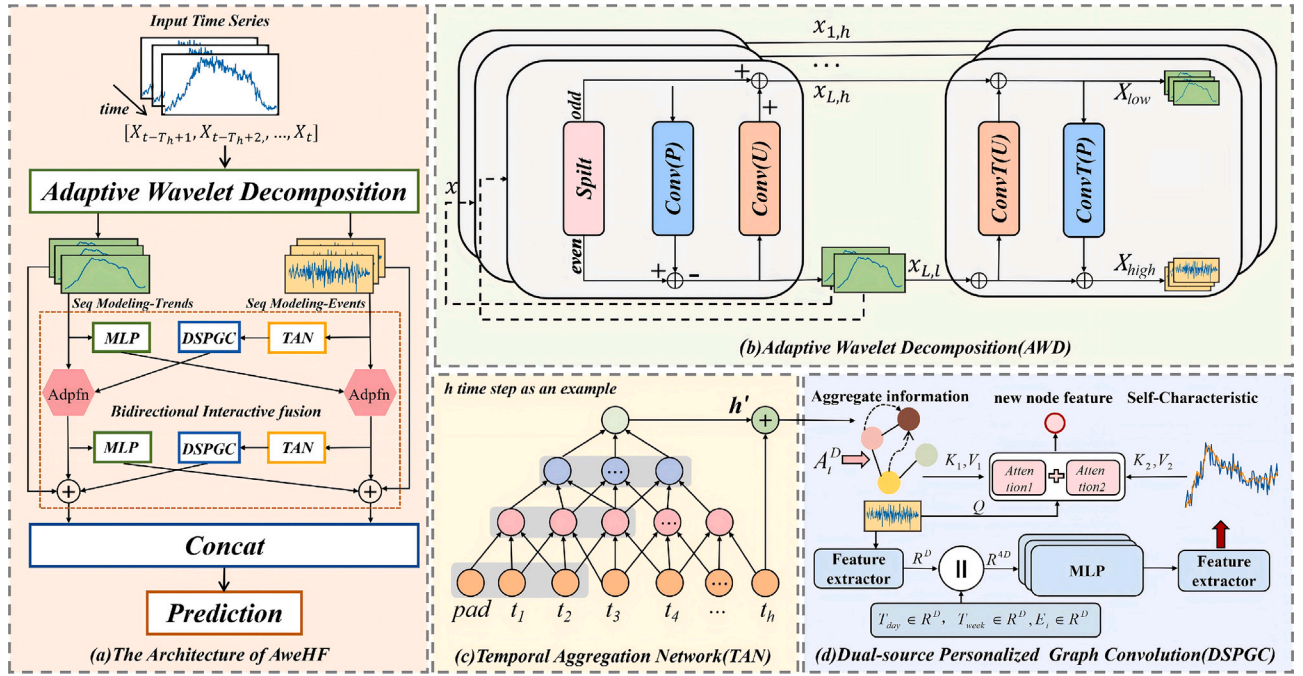


Fig. 2. The Overall architecture of AweHF. The model comprises four key components: (a) The main framework overview, (b) an Adaptive Wavelet Decomposition (AWD) module to separate trend and event components, (c) a Temporal Aggregation Network (TAN) to capture temporal dependencies, and (d) a Dual-source Personalized Graph Convolution (DSPGC) module to model spatial correlations.

variables. For example, STGCN [34] applies Graph Convolutional Networks to capture spatial correlations and combines temporal convolution for sequential modeling. DCRNN [35] employs diffusion convolution with RNN to simulate information propagation across graph structures. AGCRNN [36] leverages an adaptive graph module to learn the implicit inter-node relationships. STFGNN [21] extracts rich information from sequences through the multi-graph fusion mechanism. In recent years, Transformer-based models have shown great promise in MTS forecasting. SAN [11] reduces non-stationarity by normalizing adaptively to empower time series forecasting. Models like Informer [37] and Crossformer [38] utilize attention mechanisms to efficiently process long sequences, achieving excellent prediction performance. Lightweight MLP-based models like STID [39] introduce spatiotemporal embeddings and achieve excellent performance through a simple fully connected network.

2.2. Time series decomposition and modeling

Researchers have attempted to decompose complex sequences into simpler subcomponents to enhance forecasting performance. For example, Autoformer [40] uses moving average operations and PatchTST [41] combines traditional Transformer models with patches. It decomposes the data into subsequence blocks with different periodic lengths, enhancing forecasting accuracy. SCINet [12] uses down-sampling and cross-scale interaction to extract features at multiple time resolutions. Although these methods have shown some effectiveness, they rely on smoothing operations, which prevent the model from capturing abrupt local events. FEDformer [26] and MSGNet [42] apply the Fourier Transform to decompose the global time series features. However, they overlook the trend and event characteristics of the time series. Based on wavelet transform methods, HiLIN [28] and STWave [29] use DWT to decompose time series into trends and events. However, the fixed wavelet basis methods are not adaptive to the variations in different datasets at various times. Besides, they have limited capability to model sudden short-term sequences. Moreover, existing decomposition methods do not model the non-stationary high-frequency

components after decomposition. Most approaches use the same model architecture for processing these components. For instance, HGCN [43] enhances feature representation through graph convolution. STIDGCN [44] integrates dynamic graph convolution into an interactive learning framework. Yet, these methods still treat high-frequency sequences as ordinary sequences and do not model the suddenness of high-frequency components.

The limitations of the aforementioned methods in handling MTS motivate for this study. Traditional decomposition methods struggle to adaptively decouple data based on its intrinsic characteristics. Moreover, the modeling process after decomposition does not fully consider the heterogeneity of components, especially failing to model the instability in high-frequency events.

3. Preliminaries

3.1. Symbols and definitions

Definition 1 (Graph). Formally, a graph is denoted as $G = (V, E, A)$, where $V = \{v_1, v_2, \dots, v_N\}$ represents the set of nodes (variables), and $|V| = N$ represents the total number of nodes. E denotes the edge set, and $A \in \mathbb{R}^{N \times N}$ is the weighted adjacency matrix that describes the relationships between nodes. If an edge exists between nodes v_i and v_j , i.e., $(v_i, v_j) \in E$, then its corresponding entry in the adjacency matrix $A_{i,j} > 0$; otherwise, $A_{i,j} = 0$. The adjacency matrix A can either be predefined based on known relationships or learned from the input data.

Definition 2 (Node embedding). Node embedding is characterized as $E \in \mathbb{R}^{N \times d}$, where d denotes the embedding dimension, satisfying $d \ll N$. It provides a low-dimensional, compressed representation of nodes within the graph. By optimizing trainable parameters, these node embeddings have been widely utilized in tasks involving graph structure inference [45].

Definition 3 (Lifting scheme). The lifting scheme, also known as the second-generation wavelet method, was proposed by Sweldens [46] in

1998 to improve the adaptability of wavelet transforms in the time domain. Unlike conventional approaches, it does not rely on predefined wavelet bases. Instead, it constructs wavelets directly in the time domain through a series of “lifting” operations, making them better suited to capture local signal characteristics. Under this framework, the input sequence x is decomposed into an approximation part c and a detail part b . The entire process consists of three main steps: split, predict, and update.

- **Split:** The input signal is divided into two non-overlapping components: even (x_e) and odd (x_o) parts, defined as $x_e(n) = x(2n)$ and $x_o(n) = x(2n+1)$, where $n \in \mathbb{Z}$.
- **Predict:** In this stage, the detail component d is generated. Let $P(\cdot)$ represent the predictor, which uses $x_e(n)$ to predict $x_o(n)$. The prediction error defines the detail signal: $d(n) = x_o(n) - P(x_e(n))$, where $n \in \mathbb{Z}$.
- **Update:** The approximation component c is obtained by refining $x_e(n)$ using the detail signal $d(n)$. Let $U(\cdot)$ represent the updater. The approximation is given by $c(n) = x_e(n) + U(d(n))$, where $n \in \mathbb{Z}$.

3.2. Problem formulation

A multivariate time series is formally defined as $X = (X_{t-T_h+1}, X_{t-T_h+2}, \dots, X_t) \in \mathbb{R}^{N \times C \times T_h}$. At time step t , the value of all variables is described as $X_t \in \mathbb{R}^{N \times C}$, where $t \in [1, \dots, T_h]$. C represents the feature dimension for each variable, and T_h represents the historical window length. The core objective of MTS is to forecast future variable values using a sequence of past historical observations:

$$[X_{t-T_h+1}, X_{t-T_h+2}, \dots, X_t] \xrightarrow{f} [Y_{t+1}, Y_{t+2}, \dots, Y_{t+Q}]. \quad (1)$$

4. Method

4.1. Overview

The overall framework of AweHF is presented in Fig. 2. The model comprises an Adaptive Wavelet Decomposition (AWD) module, a sequence modeling module (high-frequency and low-frequency), and a bidirectional interactive fusion framework.

- **Adaptive wavelet decomposition:** Most existing methods focus on spatiotemporal surface decomposition and neglect the dynamic coupling between trends and events in MTS. Fixed decomposition methods struggle to adapt to the non-stationarity of different datasets. Therefore, we introduce the AWD module. This module performs Adaptive Wavelet Transform (AWT) through learnable convolutional kernels, decomposing the sequence into low-frequency and high-frequency signals. Moreover, we introduce the inverse adaptive wavelet transform (IAWT) to ensure that the decomposed frequency-domain features align with the input format of downstream models. This restores the frequency-domain signals back to the time domain, as shown in Fig. 2(b). This decomposition method effectively decouples trends and events and enhances the adaptability of the model to multiscale time series.
- **Sequence modeling module:** To accurately capture the unique properties of trend and event sequences, the model introduces the Temporal Aggregation Network (TAN), as shown in Fig. 2(c) and the Dual-Source Personalized Graph Convolution (DSPGC), as shown in Fig. 2(d), to learn spatiotemporal dependency features of high-frequency sequences. Meanwhile, a lightweight MLP is used to model the smooth features of low-frequency sequences. By processing different sequences separately, the model achieves a complementary balance between local event dynamics and global trend features.
- **Bidirectional interactive fusion framework:** Trends and events interact in MTS data. Therefore, AweHF employs a bidirectional interactive fusion strategy and introduces an Adaptive Fusion

Module, as shown in Fig. 2(a) and Fig. 4. This approach facilitates bidirectional information flow and interaction between trends and events, enhancing the ability of the model to capture cross-scale features.

4.2. Adaptive wavelet decomposition

Traditional time series decomposition methods often rely on fixed basis functions. This makes them unable to effectively adapt to changes across different datasets or within the same dataset over different time periods. Consequently, these methods tend to offer suboptimal decomposition performance. To address this issue, we use the AWD to decompose the time series. The AWD module performs AWT using learnable convolutional kernels, replacing the traditional fixed basis functions. This allows it to dynamically adapt to the characteristics of different time series. The AWT is developed based on the lifting scheme [46], and its decomposition process can be summarized in three main steps: Split, Predict, and Update, as exhibited in Fig. 2(b). Specifically, given the original time series $x \in \mathbb{R}^{C \times N \times T}$ and a predefined decomposition level L , the decomposition process at the l -th layer is as follows:

$$\begin{aligned} (x_{\text{even}}^l, x_{\text{odd}}^l) &= \text{Split}(x^l), \\ d^{l+1} &= x_{\text{odd}}^l - P(x_{\text{even}}^l), \\ c^{l+1} &= x_{\text{even}}^l + U(d^{l+1}), \end{aligned} \quad (2)$$

where $l \in \{0, \dots, L-1\}$.

The original sequence is first split into even and odd subsequences based on index parity using $\text{Split}(\cdot)$. These two subsequences are often highly correlated. For example, in many time series, consecutive points exhibit small variations or similar states. Therefore, the even subsequence x_{even} can be used to approximately predict the odd subsequence x_{odd} . Here, $P(\cdot)$ represents the predict operator and $U(\cdot)$ represents the update operator, both are implemented using learnable convolutional kernels. The predict operator $P(\cdot)$ attempts to estimate the values of the odd positions from the smoother even positions. Prediction errors are inevitable because the consecutive points are similar but not identical. We denote this error as d^l , which captures the variations that cannot be inferred from the smooth trend. In other words, d^l represents the high-frequency component of the sequence, reflecting sudden changes, local perturbations, and other high-frequency events. To ensure that any trend information not captured by $P(\cdot)$ is retained, the update operator $U(\cdot)$ feeds the average trend of the high-frequency signal back to the even subsequence. Thus, the updated sequence c^l better approximates the underlying smooth trend, forming the low-frequency component of the time series.

We apply IAWT to convert the decomposed frequency-domain signals into the time domain to make the data match the input format of subsequent spatiotemporal modeling modules and support effective spatiotemporal feature learning. The conversion formulas are as follows:

$$\begin{aligned} x_{\text{even}}^l &= c^{l+1} - U^T(d^{l+1}), \\ x_{\text{odd}}^l &= d^{l+1} + P^T(x_{\text{even}}^l), \\ x^l &= \text{Merge}(x_{\text{even}}^l, x_{\text{odd}}^l), \end{aligned} \quad (3)$$

where $U^T(\cdot)$ and $P^T(\cdot)$ are denoted learnable reconstruction operators corresponding to the decomposition stage. These operators are not required to be strict mathematical inverses, but are designed to reconstruct time-domain representations for subsequent spatiotemporal modeling. They are implemented using transposed convolutions. The $\text{Merge}(\cdot)$ operation combines the even subsequences and the odd subsequences, restoring the time series to its original length. Finally, the high-frequency event sequence $x_{\text{high}} \in \mathbb{R}^{D \times N \times T}$ is constructed from all high-frequency signals using IAWT. The low-frequency trend sequences $x_{\text{low}} \in \mathbb{R}^{D \times N \times T}$ are constructed from the low-frequency signal of the

final layer via IAWT. The corresponding formulas are as follows:

$$x_{\text{high}} = \sum_{l=1}^L \text{IAWT}(d^l), x_{\text{low}} = \text{IAWT}(c^L). \quad (4)$$

This adaptive decomposition method effectively separates trends from events, and enhances the model's ability to adapt to multiscale time series. The entire AWD module is formally defined as:

$$(x_{\text{high}}, x_{\text{low}}) = \text{AdaptiveDecompose}(x). \quad (5)$$

4.3. High-frequency sequences modeling

High-frequency event sequences contain short-term fluctuations in time series and are characterized by burstiness and instability. Some existing methods treat high-frequency sequences as ordinary sequences and do not model their specific characteristics. However, this simplified processing often results in insufficient modeling of sudden events. To address this issue, we propose the TAN (shown in Fig. 2c) and DSPGC (shown in Fig. 2d) modules to specifically model the instability of high-frequency events. Specifically, TAN first performs multiscale aggregation of high-frequency features along the time dimension. This process extracts the burst characteristics of events, achieves global information aggregation, and provides the subsequent DSPGC module with a global perception of sudden changes. Subsequently, the DSPGC module further models the personalized dynamic dependencies between nodes in the spatial dimension. With the collaboration of these two modules, AweHF can effectively capture the spatiotemporal dependencies of high-frequency events and achieve accurate modeling of high-frequency components.

4.3.1. Temporal aggregation network

In the decomposition stage, the AWD module decomposes the input time series into low-frequency trends and high-frequency events. Considering that high-frequency events exhibit short-term volatility, to address this, we design the TAN module with multiple layers of 2D-CNNs. It preserves key dynamic information of high-frequency events while providing more efficient feature input for subsequent spatial modeling. The model structure is shown in Fig. 2(c) (using an input time step h as an example). Specifically, we use the high-frequency sequence $x_{\text{high}} \in \mathbb{R}^{D \times N \times T}$ obtained from the AWD module. For each prediction step, a sliding temporal window of length h is extracted from x_{high} as the input to TAN, i.e., $x_{\text{high}}^{(t)} \in \mathbb{R}^{D \times N \times h}$. The TAN module performs convolution solely along the temporal dimension, while preserving the feature and node dimensions. For notational simplicity, we omit the superscript (t) and denote the windowed input as x_{high} in the following.

$$x_{\text{high}} * f = \sum_{k=0}^{K-1} f(k) \cdot x_{\text{high}}[\dots, t-k], \quad (6)$$

where f denotes the convolutional kernel, $*$ denotes the convolution operation, and K denotes the kernel size. The convolution kernel of size K is applied along the temporal dimension T , while the node and feature dimensions are preserved and treated as parallel channels. Through multi-layer convolution operations, the input time step h is gradually compressed to 1. This process generates an intermediate representation h' , which achieves a global information aggregation along the time dimension. In addition, since the last time step of the subsequence usually contains the latest state information, we introduce a residual connection following the convolution. Subsequently, TAN integrates the information from the last time step into the aggregated feature h' . Finally, TAN outputs the global temporal aggregated feature $H_{TAN} \in \mathbb{R}^{D \times N \times 1}$. This output provides a global feature input for the subsequent DSPGC module and significantly reduces the computational cost of subsequent spatial modeling.

4.3.2. Dual-source personalized graph convolution

After obtaining node temporal features H_{TAN} from the TAN module, we need to further analyze the complex spatial interactions of high-frequency events. The spatial dependencies of high-frequency events are dynamic and personalized. Dynamism refers to the rapid changes in spatial relationships between nodes over time. Personalization means node fluctuations are influenced by both the dynamic features of external neighbors and the node's own internal features. Traditional graph convolutional networks tend to smooth node features during neighborhood aggregation. This makes it difficult to capture the rapid fluctuations of high-frequency events. We overcome this challenge by designing the DSPGC module, as illustrated in Fig. 2(d). It uses an adaptive fusion strategy to dynamically balance the contributions from external and internal sources during each aggregation step. This preserves node-specific characteristics while effectively capturing spatial dependencies.

In DSPGC, "dual-source" refers explicitly to (i) an external source, represented by dynamically inferred neighborhood information encoded in A_t^D , and (ii) an internal source, represented by node-specific personalized features $\tilde{X}$. The core objective is to adaptively weight these two sources during graph convolution rather than relying on fixed neighborhood aggregation.

The sudden nature of high-frequency events means that the spatial relationships among nodes exhibit high dynamism. Static or predefined graph structures cannot capture such instantaneous relationships. Therefore, DSPGC begins by inferring a dynamic adjacency matrix A_t^D for the current time step, as presented in Fig. 3. This is achieved by a graph-based learning method that fuses heterogeneous information. First, given the dynamic input information $X_t \in \mathbb{R}^{N \times C}$ at time t , where C is the original feature dimension of nodes, we use CNN to capture the dynamic characteristics $X'_t \in \mathbb{R}^{N \times d}$, where d denotes the node embedding dimension. Meanwhile, a learnable node embedding $E_d \in \mathbb{R}^{N \times d}$ is initialized to encode the inherent structural correlations of nodes. We employ a GRU for adaptive fusion to balance static embeddings with dynamic features and avoid single-source biases. The corresponding formulas are as follows:

$$\begin{aligned} r_t &= \sigma(W_r E_d + W_r X'_t), \\ z_t &= \sigma(W_z E_d + W_z X'_t), \\ \tilde{h}_t &= \tanh(W_H X'_t + W_H(r_t \odot E_d)), \\ H_t &= (1 - z_t)E_d + z_t \odot \tilde{h}_t, \end{aligned} \quad (7)$$

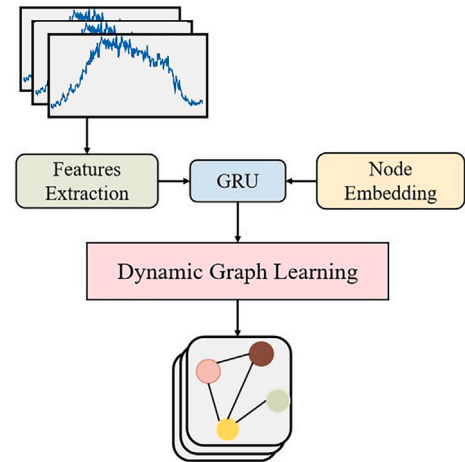


Fig. 3. The dynamic graph learning module, which learns a time-varying adjacency matrix by fusing static embeddings and dynamic features using a Gated recurrent unit (GRU).

where W_r , W_z , and W_H are learnable parameters. H_t captures the fused information at time step t . Subsequently, we use cosine similarity to compute the dynamic adjacency matrix. After that, we eliminate weak connections with $\text{ReLU}(\cdot)$ and normalize the adjacency matrix using the SoftMax function. We finally obtain the dynamic graph A_t^D :

$$A_t^D = \text{Softmax}(\text{ReLU}(H_t H_t^T)). \quad (8)$$

The internal features of nodes capture their personalized dynamic representations. In particular, they contain high-frequency fluctuation information and integrate multiple types of spatiotemporal embedding information. To model these features, we concatenate the high-frequency components x_{high} with three trainable spatiotemporal embeddings: the spatial embedding $E_s \in \mathbb{R}^D$, the daily periodic embedding $T_d \in \mathbb{R}^{N_d \times D}$, and the weekly periodic embedding $T_w \in \mathbb{R}^{N_w \times D}$. Formally, the multi-source feature representation is defined as $\hat{H} = x_{\text{high}} \parallel T_w \parallel T_d \parallel E_s$, where D denotes the embedding dimension, N_d denotes the daily sample count, and N_w denotes the number of days in a week. The concatenated features are then processed by multilayer fully connected networks to extract deep representations. To ensure training stability and efficiency, we introduce residual connections at each layer:

$$\hat{H}^{l+1} = FC_l^l(\text{ReLU}(FC_l^l(\hat{H}^l))) + \hat{H}^l, \quad (9)$$

where l denotes the current layer index. Finally, the last-layer features are projected into the target dimension to obtain the internal patterns $\tilde{X} = FC(\hat{H}_L)$ of the node. This ensures that $\tilde{X}$ preserves the unique characteristics of each node.

The external features of nodes reflect the dynamic information from neighboring nodes. Their initial representation features originate from temporal information H_{TAN} output by the TAN module, i.e., $Z^{t,0} = H_{TAN}$, where $Z^{t,0}$ denotes the aggregated external features at time step t . After obtaining the internal features $\tilde{X}$ and aggregated external features, a potential issue arises. Using only neighborhood features or only the own features of the node may cause incomplete information usage and result in performance bias. To address this issue, we introduce a dual-branch attention mechanism $\text{atten}(x_{\text{high}}, \tilde{X}, Z)$, where Z denotes the aggregated external features. This mechanism adaptively balances the contributions of neighborhood information and the node's own representation during each aggregation layer. Take node i at time t during the k -th neighborhood aggregation as an example, the high-frequency input $x_{\text{high},i}^t$ serves as the query. The internal branch uses the internal feature $\tilde{X}$ as the key and value, while the external branch uses the neighborhood feature $Z_i^{t,k}$ as the key and value. We separately compute the correlations between the input and the aggregated features. Thus, we derive the following raw attention weights:

$$\begin{aligned} \omega_{Z_i}^{t,k} &= \tanh(V x_{\text{high},i}^t + U_k Z_i^{t,k}), \\ \omega_{\tilde{X}_i}^{t,k} &= \tanh(V x_{\text{high},i}^t + W_k \tilde{X}_i), \end{aligned} \quad (10)$$

where V , U_k and W_k represent learnable transformation matrices. We apply the Softmax function to obtain weights $\phi_{Z_i}^{t,k}$ and $\phi_{\tilde{X}_i}^{t,k}$ to obtain normalized weights. A higher $\phi_{Z_i}^{t,k}$ indicates that the neighborhood aggregation feature plays a dominant role at the current moment, whereas a larger $\phi_{\tilde{X}_i}^{t,k}$ means the internal feature is dominant. Note that $\phi_{Z_i}^{t,k} + \phi_{\tilde{X}_i}^{t,k} = 1$. The calculation formula is as follows:

$$\phi_{Z_i}^{t,k} = \text{Softmax}(\omega_{Z_i}^{t,k}) = \frac{\exp(\omega_{Z_i}^{t,k})}{\exp(\omega_{Z_i}^{t,k}) + \exp(\omega_{\tilde{X}_i}^{t,k})}. \quad (11)$$

Based on dynamic weights and A_t^D , we perform multi-hop neighborhood aggregation to update node features. The node update process is formulated as follows:

$$Z^{t,k+1} = (1 - \phi_Z^{t,k}) \hat{A}_t^D Z^{t,k} + \phi_{\tilde{X}}^{t,k} \tilde{X}, \quad (12)$$

where $\hat{A}_t^D = D^{-1} A_t^D$ is a normalized dynamic graph transition matrix, and D^{-1} denotes the inverse degree matrix. After K layers of

personalized graph convolution, the multi-hop neighborhood features $\{Z^{t,0}, Z^{t,1}, \dots, Z^{t,K-1}\}$ are obtained. To fully leverage the information from different diffusion orders, we use learned weights ϕ_k to fuse the results from different aggregation steps. Finally, we obtain high-frequency features with temporal information as follows:

$$X_{\text{high}} = \sum_{k=0}^{K-1} \phi_k Z^{t,k} \quad (13)$$

where $k \in [0, K-1]$ represents the level of information diffusion. ϕ_k is used to regulate the aggregation influence of different convolutional layers. Finally, we obtain the high-frequency feature X_{high} through K convolutional layers.

4.4. Low-frequency sequences modeling

The low-frequency trend sequences generally reflect smooth patterns of time series. Compared with the complex fluctuations of high-frequency components, the low-frequency trend sequences are easier to fit [47]. Therefore, we use a lightweight processing method for the low-frequency component, whose core idea is drawn from STID [39]. Our method combines MLP with the spatial information of nodes to capture the smooth variations in low-frequency trends. The subsequent fully connected layer is denoted as $FC(\cdot)$. To explicitly incorporate static spatial correlations between nodes into low-frequency series modeling, we introduce a trainable spatial embedding matrix $E \in \mathbb{R}^{N \times d}$. This matrix provides static contextual encoding for each node and positional features for the low-frequency sequence. For each time step t , x_{low}^t obtained from the AWD module is mapped to the hidden space; that is, $H_{\text{low}}^t = FC(x_{\text{low}}^t)$, where $H_{\text{low}}^t \in \mathbb{R}^{N \times d}$ and d is the hidden dimension. Then, we fuse the spatial embedding matrix E to generate a sequence representation, yielding $\hat{H}_{\text{low}}^t = H_{\text{low}}^t \parallel E$. $\hat{H}_{\text{low}}^t$ denotes the low-frequency component matrix at time t following spatial embedding enhancement. Subsequently, we use $FC(\cdot)$ to extract the sequence features following spatial enhancement, as shown in the following formula:

$$\hat{H}_{\text{low}}^{t,l+1} = FC_l^l(\text{ReLU}(FC_l^l(\hat{H}_{\text{low}}^{t,l}))) + \hat{H}_{\text{low}}^{t,l}, \quad (14)$$

where $l \in [1, L]$ denotes the l -th layer. We introduce residual connections in each layer to strengthen the model's expressiveness. Finally, we utilize a fully connected layer to map the extracted feature $\hat{H}_{\text{low}}^{t,L+1}$, thereby generating the output X_{low} :

$$X_{\text{low}} = FC(\hat{H}_{\text{low}}^{t,L}). \quad (15)$$

4.5. Bidirectional interactive fusion

After performing spatiotemporal modeling on low-frequency and high-frequency sequences, we obtain their feature representations X_{low} and X_{high} . The low-frequency series captures long-term trends but is insensitive to sudden changes. In contrast, the high-frequency series reflects short-term dynamics but lacks global consistency. To make full use of their complementary information, we apply a bidirectional interactive fusion strategy, as shown in Fig. 2(a). This strategy enhances features through bidirectional information flow between low- and high-frequency components.

We introduce an effective **Adaptive fusion module** (AdpFn) to achieve effective information exchange between the trend (low-frequency) and event (high-frequency) components. This mechanism is designed to leverage the distinct characteristics of each sequence to improve prediction stability. Its structure is illustrated in Fig. 4. The AdpFn module serves as a dynamic interface that allows one sequence to selectively aggregate information from another using an attention mechanism. Specifically, we define two complementary interaction modes by changing the source of its Query (Q), Key (K), and Value (V):

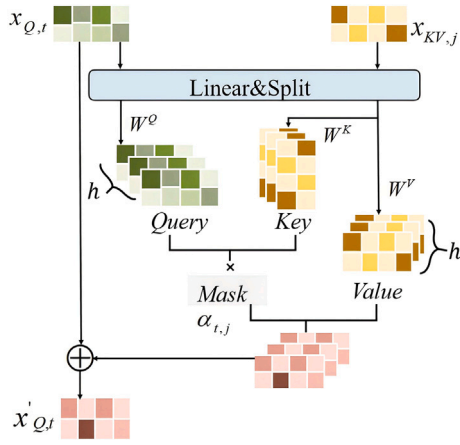


Fig. 4. The adaptive fusion module. It fuses trend and event features by projecting them into Query (Q), Key (K), and Value (V) to compute adaptive attention weights.

- **Mode 1: Trend-guides-event.** In this mode, we use the trend sequence as the Query, while the event sequence provides the Key and Value. The motivation is that the long-term trend should act as a stable guide. It queries and weights important information from the short-term event sequence. This process helps suppress excessive fluctuations caused by high-frequency noise and ensures the overall consistency of the predictions.
- **Mode 2: Event-influences-trend.** Conversely, in this mode, we use the event sequence as the Query, and the trend sequence serves as the Key and Value. This setup acknowledges that short-term, sudden events (e.g., traffic accidents) can temporarily alter or impact the long-term trend. By allowing the event sequence to query the trend background, the model can dynamically adjust its baseline forecast. This helps avoid overly smooth predictions that ignore critical dynamics.

Let X_Q denote the query sequence and X_{KV} denote the key-value sequence. The core operation of $\text{AdpFn}(X_Q, X_{KV})$ is defined as follows:

$$\begin{aligned}
 x'_{Q,t} &= \text{AdpFn}(X_Q, X_{KV}) \\
 &= x_{Q,t} + \sum_{j=1}^L \alpha_{t,j} (x_{KV,j} W^V), \\
 \text{where } \alpha_{t,j} &= \frac{\exp\left(\frac{(x_{Q,t} W^Q) \cdot (x_{KV,j} W^K)}{\sqrt{d_k}}\right)}{\sum_{k=1}^L \exp\left(\frac{(x_{Q,t} W^Q) \cdot (x_{KV,k} W^K)}{\sqrt{d_k}}\right)}
 \end{aligned} \tag{16}$$

Here, $x_{Q,t}$ is the t -th vector of the query sequence X_Q , and $x'_{Q,t}$ is its corresponding fused output. The attention weight $\alpha_{t,j}$ quantifies the importance of the j -th vector of the key-value sequence, $x_{KV,j}$, to the query vector $x_{Q,t}$. And W^Q , W^K , and W^V are learnable projection matrices that transform the inputs into query, key, and value spaces.

Finally, the output is produced by a weighted sum of the value vectors, which is then added to the original query vector $x_{Q,t}$ through a residual connection. This design enables the model to dynamically fuse features from the key-value sequence while preserving the original information of the query sequence, leading to a more robust and expressive representation.

By employing these two modes, the AdpFn module enables a dynamic, bi-directional fusion of low- and high-frequency information. This ensures that the model can capture stable long-term patterns while

also responding effectively to sudden events.

$$\begin{aligned}
 X_{\text{low}}^{\text{adp}} &= \text{AdpFn}(X_{\text{high}}, x_{\text{low}}), \\
 X_{\text{high}}^{\text{adp}} &= \text{AdpFn}(X_{\text{low}}, x_{\text{high}}),
 \end{aligned} \tag{17}$$

To avoid information loss in the adaptive fusion process, we further process both low-frequency and high-frequency components. After adaptive fusion processing, the low-frequency features $X_{\text{low}}^{\text{adp}}$ are processed by an MLP to extract long-term trend patterns. At the same time, the high-frequency features $X_{\text{high}}^{\text{adp}}$ are refined by the TAN and DSPGC modules to capture high-frequency spatiotemporal dependencies and generate X'_{high} . In this approach, low-frequency trends provide directional constraints, while high-frequency events enhance local responses. Finally, we obtain mutually fused feature representations:

$$\begin{aligned}
 X_{\text{high}}^{\text{update}} &= X_{\text{high}}^{\text{adp}} + X'_{\text{low}}, \\
 X_{\text{low}}^{\text{update}} &= X_{\text{low}}^{\text{adp}} + X'_{\text{high}}.
 \end{aligned} \tag{18}$$

4.6. Optimization objective

To alleviate potential information loss in the AWD module, we introduce residual connections outside the model. Additionally, residual connections are also used within the bidirectional interactive fusion to prevent gradient vanishing in the model. The output module of the AweHF comprises two convolutional layers as well as a ReLU activation function. The formulation is given below:

$$H = \text{Concat}\left(X_{\text{low}}^{\text{update}}, X_{\text{high}}^{\text{update}}\right), \tag{19}$$

$$\hat{Y} = \text{Conv}(\text{ReLU}(\text{Conv}(H))), \tag{20}$$

where Conv denotes a 1×1 convolution operation, and $\hat{Y}$ denotes the prediction result. We use the mean absolute error (MAE) as the optimization objective during training. The model parameters are updated by minimizing the loss between predictions and ground truth. The training criterion is formulated as:

$$\Theta = \underset{\Theta}{\text{argmin}} \frac{1}{N} \sum_{i \in N} \text{loss}(\hat{Y}_i, Y_i), \tag{21}$$

where Θ denotes the collection of AweHF parameters, and $\text{loss}(\cdot)$ is the MAE function. The entire training process of the model is summarized in [Algorithm 1](#).

5. Experiments

For the systematic evaluation of AweHF, we conducted comprehensive experiments on six real-world time series benchmarks from traffic and marine domains. The experimental design is centered around five primary research questions:

- **RQ1:** How does AweHF perform compared with the baseline methods?
- **RQ2:** How do different components of the AweHF model affect its performance?
- **RQ3:** How do key hyperparameters influence the performance of AweHF?
- **RQ4:** Do the core components of AweHF enhance predictive performance?
- **RQ5:** How does AweHF perform in real-world scenarios?
- **RQ6:** How robust and efficient is AweHF for practical deployment?

5.1. Benchmark datasets and evaluation metrics

The model is validated using six real-world datasets widely used in the traffic and oceanography domains. We provide statistical details of

Algorithm 1 Overall process of AweHF.

Input: Dataset $D \in R^{N \times T \times C}$

Parameter: $E_d, B, iter, epoch, E_s, T_d, T_w$

Output: Prediction value $\hat{Y}$

- 1: Init model parameter $\Theta, E_d, E_s, T_d, T_w$
- 2: **for** r in $1:epoch$ **do**
- 3: **for** i in $1:iter$ **do**
- 4: Sample $X \in R^{B \times N \times H}, Y \in R^{B \times N \times Q}$ from D
- 5: $A_t^D \leftarrow \{E_d, X\}$ using Eqs. (7), (8) {get a dynamic graph}
- 6: $x_{high}, x_{low} \leftarrow \{X\}$ using Eq. (5) {get high-frequency and low-frequency components}
- 7: $H_{TAN} \leftarrow \{x_{high}\}$ using Eq. (6) {capturing temporal characteristics based on high-frequency features}
- 8: $\tilde{X} \leftarrow \{E_s, T_d, T_w, x_{high}\}$ using Eq. (9) {capturing the internal characteristics}
- 9: $X_{high} \leftarrow \{\tilde{X}, A_t^D, H_{TAN}\}$ using Eqs. (10)–(13) {learning high-frequency personalized features}
- 10: $X_{low} \leftarrow \{x_{low}\}$ using Eqs. (14), (15) {learning low frequency spatiotemporal dependencies}
- 11: $X_{high}^{update}, X_{low}^{update} \leftarrow \{X_{high}, X_{low}\}$ using Eqs. (16)–(18) {high-low frequency components features interaction}
- 12: $\hat{Y} \leftarrow \{X_{high}^{update}, X_{low}^{update}\}$ using Eqs. (19), (20)
- 13: Back-propagate using Eq. (21)
- 14: $i \leftarrow i + 1$
- 15: **end for**
- 16: $r \leftarrow r + 1$
- 17: **end for**

these datasets in Table 1. The four traffic datasets PeMS03, PeMS04, PeMS07 and PeMS08 [48] are derived from the CalTrans' (California Transportation) Performance Measurement System, commonly designated as PeMS. They cover a range of traffic dynamics within the CalTrans' highway network. We obtained the other two datasets from ERA5 reanalysis data supplied by the European Centre for Medium-Range Weather Forecasts (ECMWF). They involve multi-node wave height monitoring data for the East and South China Seas.

We used the past 12 time steps to predict the next 12 time steps for all experiments. The datasets are divided into training, validation, and test sets in a 6:2:2 ratio. For the wave height dataset, although the model is trained to predict the full 12-step future sequence, we report the forecasting performance at different horizons (1, 2, 4, and 8 steps ahead) by extracting the corresponding steps from the predicted sequence, which is a common practice in ocean forecasting to evaluate short- and medium-term predictive capability. Z-score normalization was applied to standardize all input data. For model evaluation, this study employs three metrics: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Square Error (RMSE). For these metrics, lower values indicate better performance.

5.2. Baselines

We compare AweHF with baseline models to assess the performance of our proposed model. These baseline models include traditional

methods, non-decomposed spatiotemporal modeling methods, and fixed sequence decoupling methods. A concise description of each baseline is provided in Table 2.

5.3. Experimental setup

The PyTorch framework was used to implement all models in this research. All experiments were conducted in an environment equipped with an Intel i7-10400F CPU (32 GB RAM) and an Nvidia GeForce 3090 Ti GPU (24 GB RAM). We used the Adam optimizer for model training over a maximum of 200 epochs. The initial learning rate was set to 0.001, and a decay strategy was employed. Specifically, if the validation metric showed no improvement for 25 consecutive epochs, the learning rate was reduced to 0.3 times its current value. The default experimental settings were as follows: the model layer was set to 1, the number of channels to 32, the dropout rate to 0.3, the AWD kernel size to 4, and the number of decomposition levels to 1. The batch size was fixed at 64 to balance training efficiency and memory consumption. To reduce the influence of random factors on the results, each experiment was repeated three times, and the reported results represent the average performance. Comprehensive experiments were conducted on all major hyperparameters to ensure optimal model configuration.

5.4. Analysis of experimental results (RQ1)

We compare our proposed AweHF with several baseline models across six real-world datasets, using MAE, RMSE and MAPE as the evaluation metrics. The results are summarized in Tables 3 and 4. Overall, AweHF consistently outperforms all baseline models across all datasets, demonstrating superior prediction accuracy. Detailed performances across different forecasting horizons are illustrated in Fig. 5. Based on the results in Tables 2–4, we analyze the following:

Traditional statistical models like VAR and ARIMA exhibit inferior performance across all datasets. Similarly, deep learning models like LSTM, which ignore spatial dependencies, and also show weak performance. Specifically, traditional models lack generalization ability and struggle to capture nonlinear dependencies. The LSTM model without graph structures cannot model the spatial correlations between nodes. This indicates that solely relying on the temporal dimension is insufficient for MTS forecasting scenarios with strong spatial correlations.

Dynamic graphs outperform static graphs in non-decomposed models. STGNN models generally outperform traditional methods. However, there are notable differences within this group. Models with dynamic graph structures, such as STFGNN and DSTAGNN, outperform those that rely on predefined static graphs such as STGCN and DCRNN. This shows that real-world spatial dependencies are temporally dynamic. Static graph models are constrained by fixed topological structures and are unable to capture these dynamic spatial-temporal correlations. Furthermore, we compared our model with non-graph models based on the Transformer architecture (Informer, Autoformer, and PatchTST), finding that modeling spatial relationships and temporal relationships is equally important. Our AweHF model also benefits from dynamic graph mechanisms, further emphasizing the importance of dynamically modeling spatiotemporal correlations.

Decomposing original sequences into low-frequency trends and high-frequency events is effective. Models with decomposition strategies

Table 1
Detailed statistics of the six datasets.

Datasets	Samples	Sample rate	Nodes	Input size	Predicted size	Time range
PeMS03	26,208	5 min	358	12	12	2018.09.01–2018.11.30
PeMS04	16,992	5 min	307	12	12	2018.01.01–2018.02.28
PeMS07	28,224	5 min	883	12	12	2017.05.01–2017.08.31
PeMS08	17,856	5 min	170	12	12	2016.07.01–2016.08.31
East China Sea WH	30,187	6 h	90	12	1, 2, 4, 8	2003.01.01–2023.08.14
South China Sea WH	30,121	6 h	91	12	1, 2, 4, 8	2003.01.01–2023.08.14

Table 2

Description and classification of baselines. Decoupled series: **N** denotes the original sequence is not decomposed; **Fixed** denotes fixed decomposition; **Adap** denotes adaptive decomposition. Graph structure: **N** denotes no graph modeling approach is employed; **Pre** represents the use of a predefined graph; **D** denotes the adoption of dynamic graph structures. Differentiated Modeling: **N** denotes using a single model for decomposed sequences; **Y** denotes specific modeling for different components.

Methods	Description	Decouple series	Graph structure	Differentiated modeling
ARIMA [30]	Combines autoregressive and moving average components to perform forecasting on time series data.	N	N	N
VAR [31]	Models relationships among multiple variables via vector autoregressive statistical data.	N	N	N
LSTM [32]	Use memory gates in Long Short-Term Memory network to model long-term dependencies of multivariate sequences.	N	N	N
DCRNN [35]	Integrates diffused graph convolution and recurrent network for spatiotemporal dynamic dependencies.	N	Pre	N
STGCN [34]	Models spatiotemporal patterns through stacked graph convolutions and temporal convolutions for MTS prediction	N	Pre	N
AGCRN [36]	Adaptive convolutional and recurrent model that learns node's patterns and evolving relationships for spatiotemporal data.	N	Pre	N
STFGNN [21]	Captures heterogeneous spatiotemporal associations via dynamically generated graph structures and multi-strategy fusion.	N	D	N
DSTAGNN [49]	Uses dynamic graph construction and improved attention mechanisms to enhance perception of spatiotemporal features.	N	D	N
D ² STGNN [24]	Decouples spatiotemporal feature learning, models them separately, then fuses to improve robustness in dynamic environments.	Fixed	D	N
FEDformer [26]	Enhances long-sequence prediction by integrating frequency domain decomposition with sequence decomposition.	Fixed	N	N
Informer [37]	Uses ProbSparse self-attention and a generative decoder for efficient long-sequence prediction.	N	N	N
Autoformer [40]	Utilizes auto-correlation to decompose time series into trend and seasonal components, enabling effective long-term forecasting.	Fixed	N	N
DLinear [50]	A single-layer linear model that decomposes time series components and models linear relationships.	Fixed	N	N
PatchTST [41]	A Transformer model with non-overlapping patch splitting and channel-independent self-attention for long-term prediction.	Fixed	N	N
STID [39]	Encodes spatiotemporal identity through embedding layer and combines with fully-connected network for prediction.	N	N	N
SCINet [12]	Applies down-sampling convolution and interactive learning to capture features from multiple time resolutions.	Fixed	N	Y
STIDGCN [44]	Integrates dynamic graph convolution within an interactive learning architecture to model spatiotemporal dependencies.	Fixed	D	Y
STWave [29]	Decomposes traffic time series into trends and events through discrete wavelet transform, models with dual-channel encoder.	Fixed	Pre	Y
HiLIN [28]	Decomposes time series into high-frequency events and low-frequency trends and fuses through adaptive module.	Fixed	D	Y
AweHF	OUR MODEL	Adap	D	Y

Table 3

Performance on highway traffic datasets. Bold: best.

Methods	PeMS03			PeMS04			PeMS07			PeMS08		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
VAR	21.09	38.76	22.36%	23.52	37.41	17.86%	39.98	54.76	20.56%	19.12	29.73	13.06%
ARIMA	35.40	45.03	30.29%	33.60	48.63	24.22%	35.63	49.15	17.43%	30.13	42.30	22.25%
LSTM	20.63	33.55	24.95%	23.82	36.63	18.13%	24.55	38.62	10.12%	21.32	32.11	17.49%
DCRNN	15.36	26.88	15.10%	19.71	31.43	13.54%	21.22	35.46	9.13%	15.26	24.28	9.96%
AGCRN	15.43	27.15	14.76%	19.83	32.26	12.97%	23.52	37.41	17.86%	15.95	25.22	10.09%
STGCN	17.53	29.06	16.93%	21.1	33.85	13.86%	22.72	35.19	9.28%	17.11	26.85	10.12%
STFGNN	16.78	28.35	16.31%	19.85	31.89	13.03%	23.42	36.32	9.23%	16.62	26.24	10.63%
DSTAGNN	15.60	27.19	14.69%	19.31	31.46	12.70%	21.56	34.66	9.12%	15.68	24.73	9.91%
D ² STGNN	14.89	26.03	15.13%	18.21	29.80	12.30%	19.71	33.21	8.25%	14.35	24.18	9.33%
FEDformer	19.36	31.25	19.88%	22.86	35.06	16.04%	19.91	32.78	8.53%	19.55	29.31	13.58%
Informer	19.22	32.75	19.60%	22.15	36.23	14.90%	28.16	45.72	11.72%	19.97	30.55	13.06%
Autoformer	18.11	28.26	17.65%	24.36	37.05	16.88%	25.99	40.83	11.75%	19.96	31.43	12.65%
DLinear	19.69	32.25	19.70%	24.61	39.5	16.11%	23.75	34.59	9.23%	20.25	32.27	12.08%
PatchTST	18.98	30.19	17.39%	24.95	40.86	17.86%	23.68	35.61	9.95%	20.72	30.76	13.75%
STID	15.52	27.63	16.43%	18.41	29.88	12.46%	19.56	32.86	8.27%	14.42	23.69	9.37%
SCINet	14.87	24.12	14.12%	18.99	30.88	11.89%	21.63	34.32	9.01%	15.75	24.78	9.80%
STIDGCN	14.57	24.59	14.79%	18.49	29.99	12.46%	19.31	32.43	8.41%	14.16	23.75	9.21%
STWave	14.93	26.50	15.05%	18.52	30.39	12.43%	19.96	33.98	8.42%	13.51	23.40	8.90%
HiLIN	14.51	23.47	14.40%	18.22	29.79	12.29%	19.23	32.41	8.19%	13.69	23.02	8.95%
AweHF	14.35	23.10	14.23%	18.12	29.38	12.13%	18.83	31.98	7.86%	13.40	22.75	8.75%

Table 4
Performance on sea wave height prediction datasets. Bold: best.

Dataset	Timestep	Metrics	Methods						
			LSTM	STGCN	STID	STIDGCN	FEDformer	HiLIN	AweHF
South China Sea WH	6-Hours	MAE	0.1579	0.1583	0.1051	0.0963	0.1475	0.0901	0.0788
		RMSE	0.2580	0.2420	0.1689	0.1539	0.2125	0.1440	0.1328
		MAPE	9.26%	8.83%	7.26%	6.41%	7.85%	6.06%	5.85%
	1-Day	MAE	0.3936	0.385	0.3096	0.3101	0.3090	0.2482	0.2385
		RMSE	0.5680	0.5505	0.4581	0.4843	0.5240	0.4013	0.3713
		MAPE	20.10%	19.06%	17.56%	19.75%	18.21%	16.09%	15.88%
	2-Days	MAE	0.5747	0.5510	0.4785	0.4638	0.5112	0.4199	0.4025
		RMSE	0.8080	0.7548	0.6936	0.7138	0.7353	0.6560	0.6367
		MAPE	33.78%	31.82%	28.99%	30.88%	30.10%	27.53%	26.74%
East China Sea WH	6-Hours	MAE	0.2531	0.2119	0.1653	0.1476	0.1926	0.1471	0.1441
		RMSE	0.3670	0.3287	0.2646	0.2450	0.2921	0.2489	0.2386
		MAPE	10.25%	10.06%	9.50%	9.16%	9.78%	8.98%	8.83%
	1-Day	MAE	0.5363	0.5130	0.4640	0.4509	0.4955	0.3940	0.3887
		RMSE	0.8121	0.7951	0.6909	0.7180	0.7325	0.6596	0.6086
		MAPE	29.13%	27.91%	25.50%	27.97%	26.75%	24.63%	23.08%
	2-Days	MAE	0.6630	0.6466	0.5812	0.5457	0.6206	0.5259	0.5002
		RMSE	0.975	0.921	0.867	0.910	0.899	0.823	0.727
		MAPE	45.06%	42.67%	36.82%	35.31%	38.31%	33.23%	31.97%

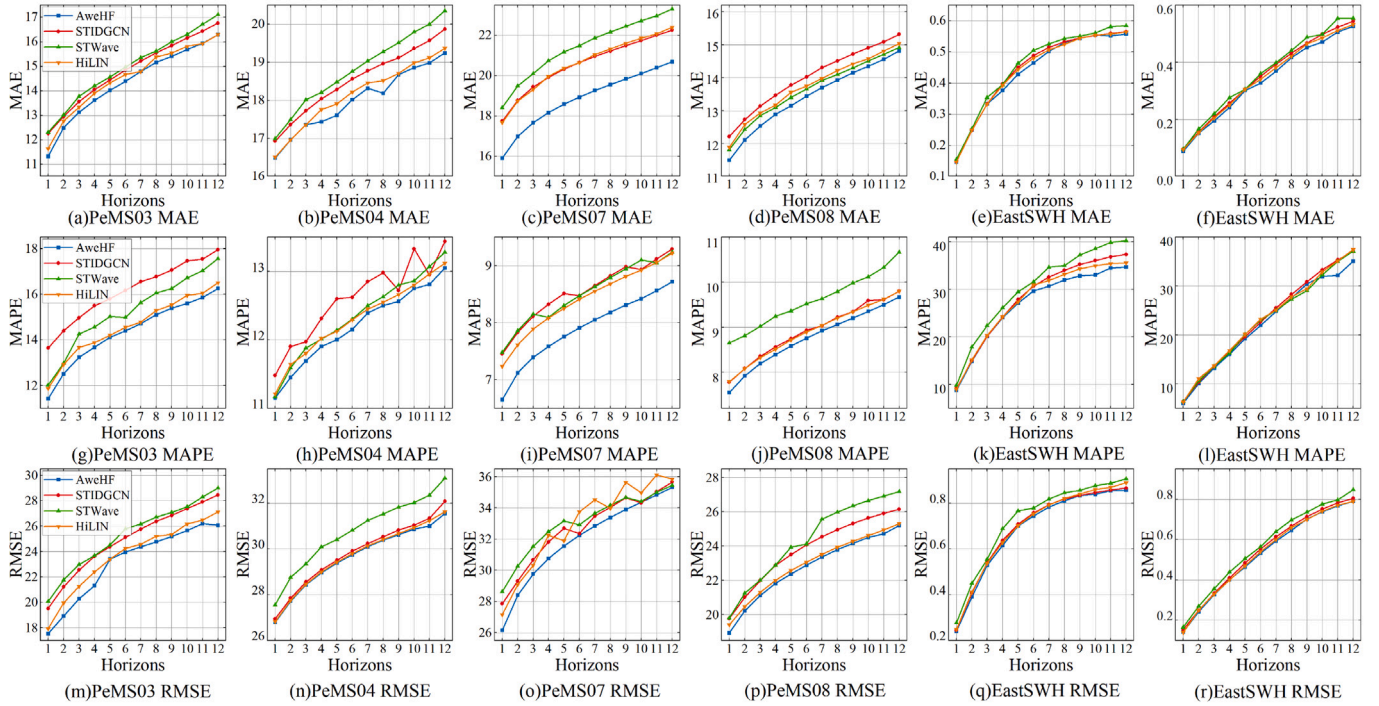


Fig. 5. Experimental prediction results of AweHF for each horizon on PeMS03, PeMS04, PeMS07, PeMS08, EastSWH and SouthSWH.

consistently outperform those without. This confirms that decoupling complex and non-stationary time series into more manageable components is useful. However, there are key differences based on the decomposition strategy. For example, SCINet captures multiscale temporal features through downsampling, while STIDGCN attempts decomposition through subsequence partitioning. But they overlook the trends and event characteristics in the time series. Models such as FEDformer, HiLIN, and STWave apply fixed decomposition methods to time series, which improve forecasting performance. However, their fixed decomposition patterns cannot adapt to variations across different datasets. Moreover, they treat high-frequency sequences as ordinary sequences, ignoring their distinctive characteristics, which leads to suboptimal forecasting performance.

In contrast, AweHF achieves the best results across all datasets and metrics. This is primarily owing to two key features: First, the AWD module performs data-driven decomposition, effectively separating trends and events. Second, the model applies differentiated strategies to capture diverse sequence characteristics, with a particular focus on personalized modeling for unstable high-frequency components. The combination of decomposition-based learning and personalized modeling of diverse sequence features enables AweHF to better handle complex non-stationary time series and significantly improve forecasting performance.

5.5. Ablation analysis (RQ2)

To further validate the effectiveness of our model's key components, we conduct ablation experiments on six datasets in this section. Five model variants are designed by removing or replacing specific modules:



Fig. 6. Performance comparison of the ablation study for AweHF and its variants across all datasets.

- **w/o AWD:** In this variant, the AWT and IAWT operations are removed and the original time series is directly input into the downstream model.
- **w/o TAN:** This variant uses a simple linear layer to replace the TAN module.
- **w/o DSPGC:** In this variant, the DSPGC in high-frequency component modeling is replaced with a standard diffusion graph convolution.
- **w/o MLP:** In this variant, the MLP operation is removed for low-frequency trends.
- **w/o DG:** In this variant, the dynamic graph learning module is replaced with a predefined static graph structure.

As shown in Fig. 6, the results clearly demonstrate the contribution of each component. The figure compares the models in terms of (a)MAE, (b)MAPE, and (c)RMSE. A detailed analysis is provided below.

The complete AweHF model consistently outperformed all its variants across the six datasets. This result indicates that each component provides a meaningful contribution to the model's forecasting accuracy. The specific contributions are detailed below.

A notable performance degradation was observed in the “w/o AWD” variant. This suggests that raw time series contain coupled trend and event components, whose interference complicates the modeling process. The Adaptive Wavelet Decomposition (AWD) module is therefore essential. It effectively decouples these components, providing a structured foundation for subsequent analysis.

The “w/o TAN” variant, which uses a simple linear layer, also shows a substantial drop in performance. The Temporal Aggregation Network (TAN) is designed to capture bursty patterns at multiple scales, providing a global awareness of high-frequency events. Without it, the model cannot supply the necessary temporal context to the graph convolution module. This finding supports the need for a specialized aggregator to process unstable, high-frequency data.

Furthermore, replacing the proposed DSPGC with a standard graph convolution (“w/o DSPGC”) also degrades performance. Standard GCNs tend to smooth node features, which can obscure local fluctuations and personalized patterns. In contrast, the dual-branch mechanism in DSPGC is designed to preserve these node-specific dynamics. The result validates the benefits of this personalized approach to graph convolution.

Finally, the remaining variants also demonstrated the importance of their respective components. The “w/o DG” variant, which employs a static graph, performed poorly, confirming that spatial correlations in the data are dynamic. The “w/o MLP” variant's underperformance shows that a simple MLP is an effective and necessary component for modeling the smooth, global characteristics of the low-frequency trend.

In summary, the ablation studies provide robust evidence for our design choices. The results confirm that the decomposition-based framework, combined with specialized modules for modeling distinct sequence features, is crucial for achieving high performance in multivariate time series forecasting.

5.6. Hyperparameter analysis (RQ3)

This section conducts a sensitivity analysis of seven key hyperparameters in the AweHF model. Specifically, the number of layers in the bidirectional interaction fusion module, the number of channels, and the dropout rate are set to $\{1, 2, 3\}$, $\{8, 12, 24, 32, 64\}$, and $\{0.2, 0.3, 0.4\}$, respectively. For the AWD module, the number of decomposition levels and kernel sizes are set to $\{1, 2, 3\}$ and $\{4, 5, 6\}$. For the MLP module, which processes low-frequency components, the MLP dimension and the number of MLP layers are set to $\{8, 16, 24, 32, 64\}$ and $\{1, 2, 3\}$. The experimental results are shown in Figs. 7 and 8. Based on these results, the following observations can be made:

First, as shown in Fig. 7(a–c), the optimal number of channels varies across datasets because each dataset has distinct characteristics. When the number of channels reaches 32, the model achieves near-optimal performance. Dropout helps improve generalization by reducing dependence on specific training samples. The results indicate that a dropout rate of 0.3 yields the best performance, as presented in Fig. 7(d–f). The impact of model depth is analyzed by varying the number of layers in the bidirectional interaction fusion framework. As described in Section 2.2, this framework integrates features learned from different modules by allowing mutual information exchange between sequences. As shown in Fig. 7(g–i), the single-layer model already achieves strong performance. Adding more layers does not bring further improvement but increases training time and the risk of overfitting. These results confirm the efficiency and effectiveness of the proposed bidirectional fusion design.

Second, for the MLP module handling low-frequency sequences, the model performs best when the MLP dimension matches the number of channels. Furthermore, when the number of MLP layers is set to three, the model achieves the highest accuracy. This suggests that an appropriate network depth is crucial for capturing the temporal dependencies of low-frequency components.

Finally, the analysis of the AWT module shows that a single-level decomposition consistently provides the best performance, as shown in Fig. 8(a–c). Although deeper decomposition can capture broader low-frequency trends, it may also weaken short-term dynamics and reduce overall prediction accuracy. For kernel size, the optimal choice depends on the dataset characteristics. For example, the PeMS03

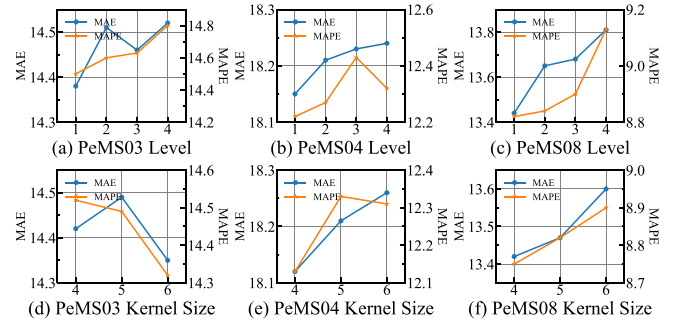


Fig. 8. Hyperparameter analysis results of AWD parameters on PeMS03, PeMS04 and PeMS08 datasets.

dataset, which contains additional nodes and higher structural complexity, benefits from a larger kernel that captures long-range dependencies, as presented in Fig. 8(d). In contrast, the PeMS08 dataset, with more rapid fluctuations, performs better with a smaller kernel that can accurately model short-term variations as illustrated in Fig. 8(f).

5.7. Study on key components in AweHF: adaptive wavelet and event-aware modeling (RQ4)

5.7.1. Analysis of adaptive wavelet decomposition

To explore the role of the AWD in MTS forecasting, we design four model variants. As introduced in Section 4.2, the AWD module decomposes the original sequence into low-frequency trends x_{low} and high-frequency events x_{high} using AWT and IAWT. These two components are then fed into downstream models for sequence processing. Based on this, we construct four variants of AweHF to study the role of AWD from three perspectives: the necessity of decomposition, the complementarity of components, and the advantages of adaptiveness. The variants are defined as follows:

- “Origin_Only”: The AWD module is removed and the original time series is directly input into the downstream model.

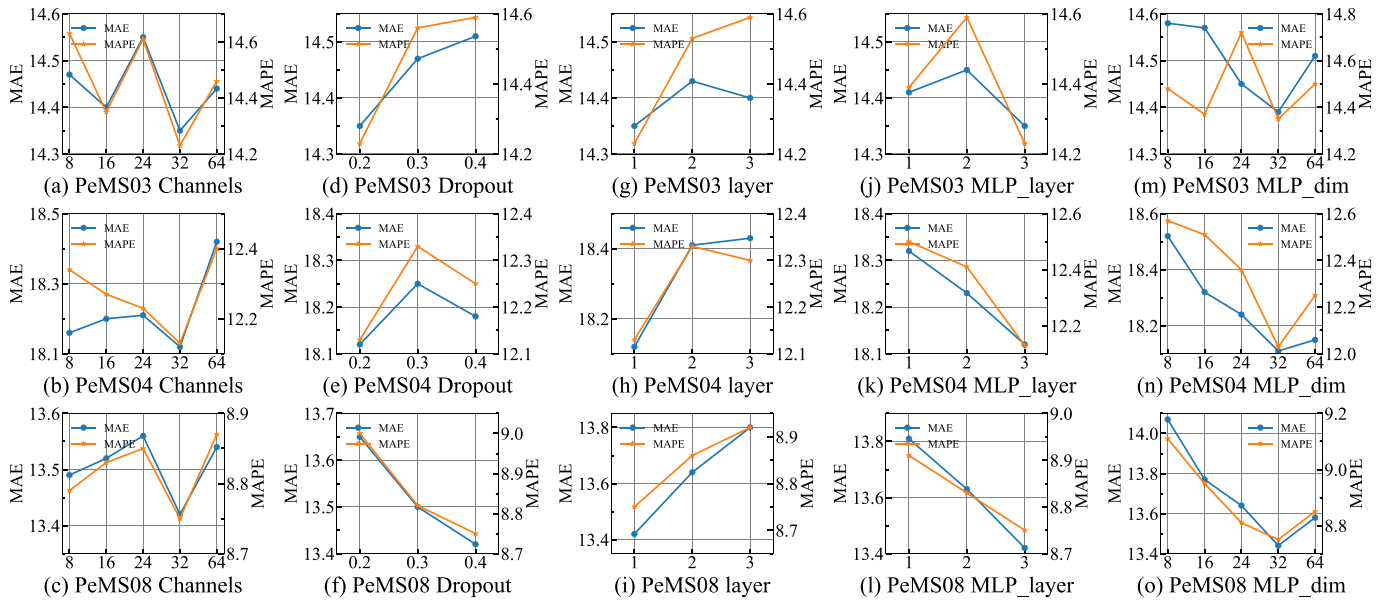


Fig. 7. Experimental results of hyperparameter analysis on PeMS03, PeMS04 and PeMS08 datasets.

- “Low-Freq_Only”: Uses only x_{low} as input into the downstream model.
- “High-Freq_Only”: Uses only x_{high} as input into the downstream model.
- “DWT_based”: Adopts the DWT to decompose the original sequence into x_{low} and x_{high} , which are then input into the downstream model.

The experimental results are reported in Table 5, and we visualize the prediction curves and their corresponding ground truth values in Fig. 9. Based on these experiments, we summarize three main findings:

- **Decomposition of time series is necessary.** As shown in Fig. 9(a) and (e), the performance of “Origin_Only” is significantly worse than decomposition-based methods. Without decomposition, long-term trends and short-term events remain coupled. Trend information cannot provide stable guidance, and event signals are weakened by interference. This results in mediocre predictions. This result is consistent with our assumption that raw sequences suffer from dynamic entanglement, and decomposition is required for effective modeling.
- **Low-frequency and high-frequency components are complementary.** As presented in Table 5, both the “Low-Freq_Only” and

“High-Freq_Only” variants perform worse than the full model. Fig. 9(b) and (f) demonstrate that using only x_{low} as input enables the model to capture overall trends but fail to detect local abrupt changes. In contrast, using only the high-frequency component x_{high} as input reflects local fluctuations but lacks global stability, as shown in Fig. 9(c) and (g). This confirms that the two components are complementary: the trend provides directional stability, while the event sequence enhances short-term sensitivity.

- **Adaptive decomposition outperforms fixed decomposition.** As illustrated in Fig. 9(d) and (h), “DWT_based” performs better than “Origin_Only” but still falls short of AweHF. This indicates that fixed DWT reduces mixed-frequency interference to some extent. However, its predefined decomposition mode cannot adapt to the characteristics of different datasets and lacks adaptability to the current task. By contrast, AWD dynamically decouples sequences in a data-driven manner, enabling more flexible and effective decomposition.

In addition, we conducted further experiments to compare different fixed wavelet bases, and the results are shown in Table 5. Based on the results it is evident that different wavelet bases (such as db1, sym2, and coif1) exhibit varying performance on the same dataset. For example, db1 outperforms sym2 and coif1 on the PeMS08 dataset. However, the same wavelet base shows inconsistent results on different datasets. For instance, db1 performs much better across PeMS08 than on PeMS03. These findings reveal that fixed wavelet decomposition lacks adaptability across datasets. In contrast, AWD adaptively adjusts the decomposition strategy according to the properties of the input sequence. This prevents suboptimal predictions caused by an unsuitable fixed wavelet basis. These results support our earlier assumption that fixed wavelet decomposition is limited in MTS forecasting.

To better understand the internal behavior of the AWD module, we analyze the learned convolutional kernels of the prediction (P) and update (U) operators. We visualize the averaged temporal impulse responses across channels to reveal the common filtering patterns learned by AWD, as shown in Fig. 10.

Table 5
Effects of different decomposition strategies on AweHF performance across PeMS03 and PeMS08. (cf. Fig. 9).

Methods	PeMS03			PeMS08		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE
Origin_Only (x)	14.53	24.27	14.80%	13.69	22.96	8.96%
Low-Freq_Only (x_{low})	14.53	23.69	14.59%	13.58	23.92	8.98%
High-Freq_Only (x_{high})	14.65	23.92	14.79%	13.63	23.93	8.97%
DWT_based (db1)	14.52	23.84	14.50%	13.53	22.86	8.90%
DWT_based (sym2)	14.46	23.21	14.68%	13.65	23.23	8.97%
DWT_based (coif1)	14.51	23.78	14.69%	13.61	22.95	8.96%
AweHF	14.35	23.10	14.23 %	13.40	22.75	8.75 %

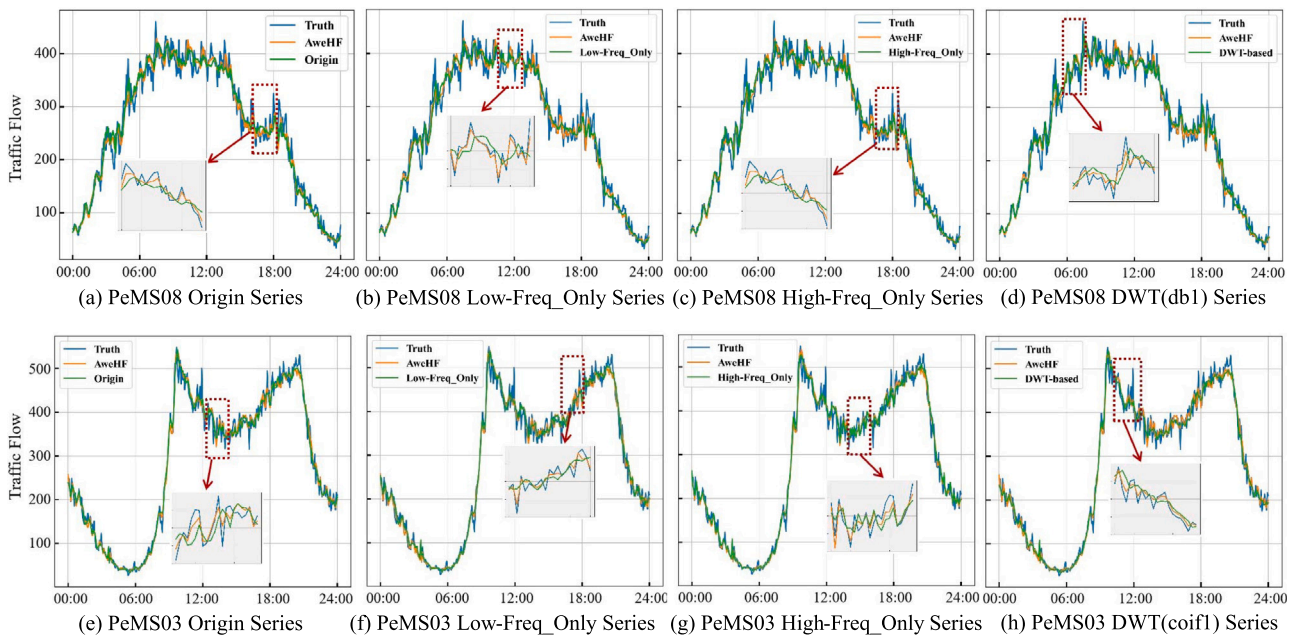


Fig. 9. Prediction visualization of AweHF under different decomposition strategies on PeMS03 and PeMS08.

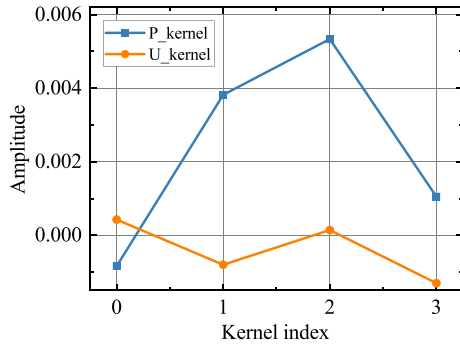


Fig. 10. Visualization of the learned convolutional kernels in the AWD module.

Table 6

Normalized error and signal reconstruction quality comparison for evaluating event-aware high-frequency modeling.

Methods	PeMS03	PeMS04	PeMS08
RRMSE (%)			
STIDGCN	13.67	13.77	10.29
STWave	15.35	13.91	10.11
HILIN	13.89	13.93	10.03
AweHF	13.38	13.49	9.83
PSNR (dB)			
w/o TAN-DSPGC	37.51	29.50	31.18
AweHF	37.90	29.55	31.26

The P kernels exhibit dominant central weights, performing local weighted averaging and producing smooth, low-pass predictions that capture slowly varying trends. In contrast, the U kernels show alternating positive and negative weights, resembling discrete differential operators that amplify rapid temporal changes while suppressing low-frequency components.

This analysis indicates that AWD adaptively learns complementary low- and high-frequency operators in a fully data-driven manner, enhancing the interpretability of the proposed adaptive wavelet decomposition.

5.7.2. Analysis of event-aware high-frequency modeling

Standard metrics like MAE, RMSE, and MAPE measure point-wise errors in time series forecasting. However, they are not sufficient to fully evaluate a model's performance on high-frequency events. These metrics have two main limitations. First, they are sensitive to the scale of the data. Second, their values are often dominated by low-frequency trends. As a result, they can obscure how well a model captures sudden, large changes in the data. To address these issues, we introduce two additional metrics to better evaluate our proposed TAN-DSPGC module: the Relative Root Mean Square Error (RRMSE) and the Peak Signal-to-Noise Ratio (PSNR).

RRMSE is defined as the standard RMSE normalized by the average value of the observed signal, as shown in Eq. (22). By normalizing the error magnitude, RRMSE becomes scale-invariant, allowing fair comparisons across datasets with different physical units or value ranges [51]. This property is particularly important in our experiments, which involve heterogeneous data such as traffic flow and wave height. As shown in Table 6, AweHF consistently achieves lower RRMSE values than other methods, indicating that its performance gains are not merely due to data scale, but reflect a genuine improvement in prediction accuracy.

$$\text{RRMSE} = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2}}{\frac{1}{N} \sum_{i=1}^N Y_i} \quad (22)$$

To specifically analyze the model's ability to handle abrupt events, we further report the Peak Signal-to-Noise Ratio (PSNR) of the predictions, defined in Eq. (23). PSNR measures the ratio between the maximum possible signal power and the power of the reconstruction error, and is highly sensitive to large deviations that typically occur during sharp and sudden changes. This makes PSNR particularly suitable for evaluating the modeling quality of high-frequency event components. We compare the PSNR values obtained by the full AweHF model and a variant without the TAN-DSPGC module (AweHF w/o TAN-DSPGC) to isolate the contribution of event-aware high-frequency modeling, as shown in Table 6.

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right), \quad (23)$$

where MAX represents the maximum value of the ground-truth signal in the corresponding dataset.

5.8. Study on prediction results and dynamic graph evolution (RQ5)

To evaluate the performance of AweHF in real-world forecasting tasks and to examine its interpretability, this section presents two case studies in the domains of traffic flow and wave height. By incorporating real-world scenario characteristics, we demonstrate that the model effectively captures spatiotemporal dependencies aligned with domain-specific patterns.

5.8.1. Analysis of traffic flow prediction results

For traffic forecasting, we selected Nodes 18 and 3 from the PeMS08 dataset, as they play a key role in the traffic flow propagation in the real-world road network. Node 18 mainly reflects cross-regional commuting traffic flow, while Node 3 primarily captures local short-distance trips. We visualized the predicted and actual values for both nodes, as shown in Fig. 11(a) and (b). From the figures, it can be observed that AweHF can accurately capture traffic flow at different nodes in various regions. This indicates that the model can effectively learn and predict complex traffic patterns.

We also compared our model with HiLIN quantitatively on these two nodes. Table 7 shows the detailed error metrics. The results indicate that AweHF outperforms HiLIN on both nodes. The performance gain is especially clear on Node 18, which has more complex traffic patterns. Here, AweHF reduces the MAE by 3.63%. This confirms our model's strength in handling challenging non-stationary data. Overall, the quantitative results support our visual analysis and prove the high accuracy of our model.

To verify the efficacy of AweHF's dynamic graph learning method, we selected the first 30 nodes for analysis. We then examined the dynamic adjacency relationships learned by the model during peak and off-peak periods, as shown in Fig. 11(c). The results show that the adjacency matrices learned in two different peak periods are highly similar. In contrast, the connections between nodes become much weaker during off-peak hours. These changes in graph structure within the same spatial region indicate that the model can dynamically capture spatiotemporal dependencies between nodes.

To further reveal AweHF's ability for personalized modeling, we visualized the dynamic changes in the internal feature weights for Nodes

Table 7

Forecasting performance on specific nodes for the PeMS08 case study (cf. Fig. 11).

Node	Metric	Methods		Improvement
		HiLIN	AweHF (Ours)	
Node 18	MAE	12.12	11.68	+ 3.63%
	RMSE	26.31	25.42	+ 3.38%
Node 3	MAE	18.87	18.50	+ 1.96%
	RMSE	25.86	25.20	+ 2.55%

18 and 3. These weights reflect the model's reliance on the historical patterns of the nodes. As observed in Fig. 11(d), we find that both nodes exhibit significant fluctuations in their internal weights. The weights are higher during the off-peak hours, such as 4 to 6 a.m. and 8 to 12 p.m. This indicates that during these time periods, the model mainly relies on the node's own historical patterns. On the other hand, during the peak hours (from 7 to 10 a.m. and from 5 to 7 p.m.), the weights decrease. This indicates that during these periods, the model is more reliant on the information from neighboring nodes. This phenomenon is consistent with real traffic patterns: during off-peak periods, traffic flow is lower, and the node's own pattern dominates. During peak hours, traffic congestion increases, and the interactions between nodes become the primary factor. This confirms that our proposed model can capture and adapt to dynamic traffic patterns at different times of the day and can achieve adaptive, personalized modeling for each node.

In addition, to provide deeper insight into how our model interprets and separates different temporal components, we further visualize in Fig. 11(e) the decomposed low- and high-frequency sequences for Node 18. It can be seen that the low-frequency component captures the

overall trend of traffic flow, while the high-frequency component reflects short-term fluctuations. This clear separation indicates that the model can effectively distinguish between trends and variations in the time series, which helps improve both the accuracy and interpretability of the predictions.

5.8.2. Analysis of wave forecast results

To evaluate the generalization capability of the model across domains, we applied it to predict sea wave height in ocean science. We used multi-node monitoring data from the South China Sea, covering the period from June 30, 2020, to June 30, 2021. As the dataset is sampled at 6-hour intervals, it effectively captures long-term trend variations. Fig. 12(a) shows that AweHF achieves excellent accuracy in predicting wave heights in the South China Sea. The predicted curves closely match the ground truth, demonstrating the model's ability to capture both long-term trends and short-term variations.

We further analyzed the learned spatial dependencies to assess whether the model captures physical ocean processes, as shown in Fig. 12(b). During winter, the South China Sea is mainly influenced

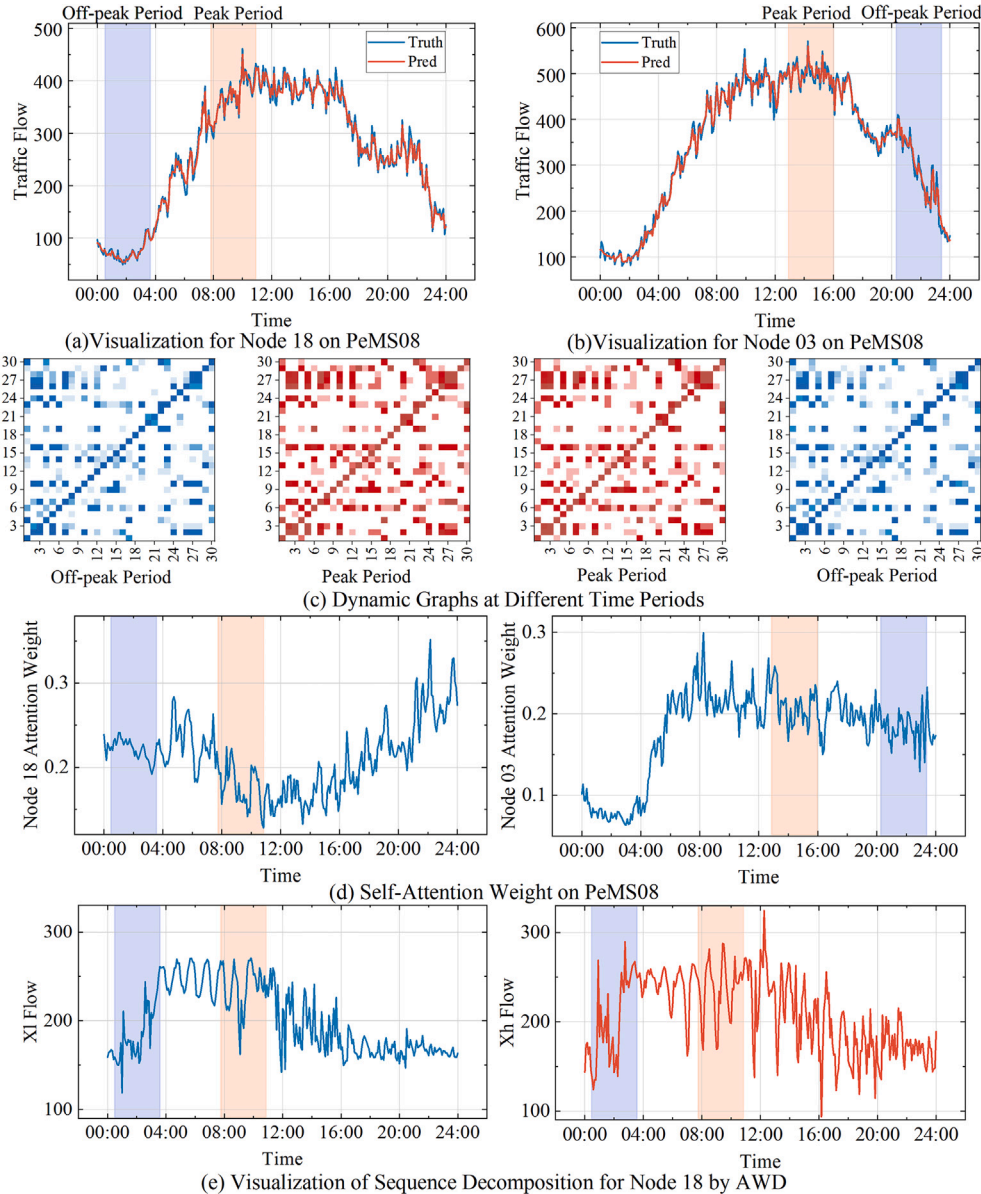


Fig. 11. Case study on traffic flow forecasting for the PeMS08 dataset.

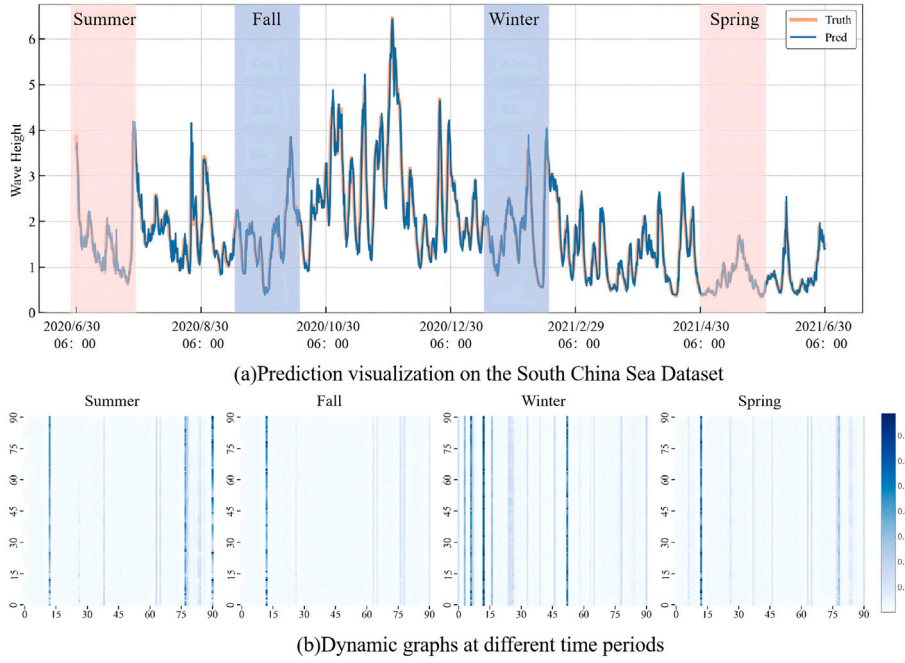


Fig. 12. Case study of wave height prediction on south china sea datasets.

by the northeast monsoon, resulting in intense wave fluctuations. The learned graph shows high heterogeneity and strong node correlations. In spring, as the monsoon weakens, wave activity stabilizes, and node correlations decrease. In summer, ocean conditions are relatively stable, and the model learns spatial correlations of uniform intensity. When autumn arrives, the northeast monsoon forms and intensifies. The correlation strength between nodes becomes stronger than in summer, and ocean waves shift from a stable state to strong fluctuations again.

These seasonal differences in graph structures show that AweHF can dynamically adapt its spatial modeling to temporal changes. The AweHF model exhibits a strong capacity to capture spatiotemporal dependencies across domains, from consistent global patterns during stable periods to localized dynamics during transitions.

5.9. Study on the applicability and efficiency of AweHF (RQ6)

5.9.1. Analysis of time complexity

The time complexity of the main modules in AweHF is summarized in Table 9, where N denotes the number of nodes, D denotes the hidden feature dimension, and L denotes the input sequence length. Specifically, AWD mainly consists of two 2D convolutions with a time complexity of $O(D^2N)$, which simplifies to $O(N)$ when D is constant. The TAN module utilizes temporal convolutions, resulting in $O(D^2N)$, reducing to $O(N)$ for constant D . DSPGC is dominated by graph convolution, resulting in $O(DN^2)$, and becomes $O(N^2)$ when D is constant. The dynamic graph learning module involves several operations with an overall complexity of $O(DN + ND^2 + N^2D)$, where $O(N^2D)$ is generally the dominant term, making the approximate complexity $O(N^2)$.

In addition to analyzing the time complexity of each module in AweHF, we also compare it with traditional wavelet transforms and leading time series models. First, the traditional wavelet transform has a time complexity of $O(N)$, which is comparable to the AWD module in AweHF. However, AWD enables adaptive decomposition, allowing automatic parameter adjustment for heterogeneous data and providing better support for complex tasks. Second, compared with dynamic GNNs such as HiLIN and STIDGCN, the overall time complexity of AweHF is also $O(N^2)$. Benefiting from the combination of AWD and TAN-DSPGC,

Table 8

Computation needs comparison of AweHF on PeMS07 and PeMS08.

Datasets	Nodes(N)	Methods	Memory usage (Training/Inference)	Time (Training/Inference)	MAE
PeMS08	180	STWave	5590M/5590M	37s/7.9s	13.51
		HiLIN	4814M/4814M	31s/7.5s	13.69
		AweHF	3840M/3840M	25s/5.4s	13.40
PeMS07	883	STWave	10587M/10587M	140s/30s	19.96
		HiLIN	9380M/9380M	143s/33s	19.23
		AweHF	9326M/9326M	59s/6.0s	18.83

AweHF improves the flexibility of trend and event separation, while enhancing the precise modeling of sudden high-frequency events. This results in better representational power and broader applicability than conventional approaches. Notably, owing to the complexity of graph structures and dynamic relationships, most spatio-temporal graph neural network models have a time complexity of at least $O(N^2)$. Finally, Transformer-based models, such as Informer, exhibit increasing complexity with increasing sequence length L , which often limits their efficiency when handling long sequences or high-dimensional data. In contrast, AweHF strikes a practical balance between efficiency, performance, and interpretability by integrating temporal aggregation (TAN) and adaptive graph learning (DSPGC). Overall, AweHF demonstrates competitive advantages by enhancing the modeling capacity for diverse time series data while maintaining computational efficiency and innovation.

Table 8 presents the training time, inference time, and memory usage of different methods on the small-scale dataset PeMS08 and the large-scale dataset PeMS07. For each dataset, all models were evaluated under the same experimental settings. Specifically, the batch size and hidden channel dimension were set to 32 for PeMS08 and 16 for PeMS07, while all other hyperparameters were kept identical across methods. The reported memory usage corresponds to the peak GPU memory consumption, and the training and inference times are measured as the average wall-clock time per epoch and per forward pass, respectively.

Table 9
Time complexity comparison analysis of AweHF and other models.

Key modules/Models	Actual time complexity	Approximate time complexity
Temporal aggregation network (TAN)	$O(N D^2)$	$O(N)$
Dual-source personalized graph convolution (DSPGC)	$O(N^2 D)$	$O(N^2)$
Dynamic graph learning	$O(N^2 D + N D^2 + N D)$	$O(N^2)$
Adaptive wavelet decomposition (AWD)	$O(N D^2)$	$O(N)$
Tranditional wavelet transformer	$O(N D)$	$O(N)$
GNN-based model(Dynamic Graph)		
HiLIN(2025)	$O(N^2 D + N D + N^2 D^2)$	$O(N^2)$
STIDGCN(2025)	$O(N^2 D + N D + N^2 D^2)$	$O(N^2)$
Transformer-based Model		
Informer	$O(L N D + L D \log L)$	$O(L N + L \log L)$
AweHF	$O(N^2 D + N D^2)$	$O(N^2)$

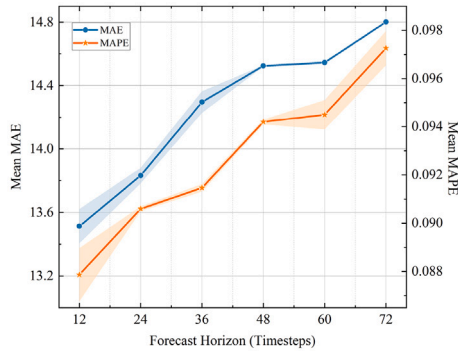


Fig. 13. Performance of AweHF on the PeMS08 dataset over different forecast horizons.

5.9.2. Analysis of model stability

To test the stability of the AweHF model, we conducted repeated experiments using multiple different random seeds. As shown in Fig. 13, this figure illustrates the average performance (solid line) and the range of fluctuations in the results across multiple runs (shaded area).

First, the shaded area in the figure is very narrow, indicating that the AweHF model exhibits highly consistent performance under different random seeds. Quantitatively, the standard deviations of MAE and MAPE across the five runs remain very small at all prediction horizons, demonstrating low variance and strong robustness to random initialization. This indicates that our model is highly reliable and reproducible.

Second, as the prediction time increases from 12 steps (1 h) to 72 steps (6 h), both the MAE and MAPE curves show very gradual increases without sharp rises. Moreover, the variance across different runs does not amplify with longer horizons. This demonstrates that our model effectively suppresses error accumulation in long-term predictions, exhibiting excellent long-term predictive capabilities.

5.9.3. Analysis of model limitations

Although AweHF demonstrates superior performance in modeling highly non-stationary spatiotemporal data, a comprehensive evaluation also requires an objective discussion of its practical limitations. These limitations primarily stem from the computational cost introduced by dynamic graph modeling and the task-oriented specialization of its architectural components.

Price of Dynamic Graphs

AweHF incorporates dynamic graph learning and dynamic graph convolution modules to explicitly model time-varying spatial dependencies. While this design significantly enhances representational flexibility, it inevitably introduces a quadratic computational overhead with respect to the number of nodes. Specifically, both the time and memory

complexity of a single inference step scale as $O(N^2)$, where N denotes the number of nodes in the traffic network.

This quadratic cost may pose scalability challenges when deploying AweHF on large-scale urban networks with thousands of sensors, especially in hardware-constrained environments. To empirically reflect this behavior, Table 8 reports the training time, inference time, and memory consumption on both a small-scale dataset (PeMS08 with 180 nodes) and a large-scale dataset (PeMS07 with 883 nodes). As expected, the computational cost increases substantially with the number of nodes, which is consistent with the theoretical $O(N^2)$ complexity analysis in Section 5.9.1.

It is worth noting that this limitation is not unique to AweHF. Most state-of-the-art spatio-temporal graph neural networks with dynamic adjacency modeling, such as HiLIN and STIDGCN, exhibit similar quadratic complexity. Compared with these representative baselines, AweHF achieves notably lower prediction error while maintaining comparable or even reduced inference time, indicating a favorable performance–efficiency trade-off in practice.

From a model design perspective, several strategies could be explored to mitigate the quadratic cost in future work. These include node sampling modules, hierarchical or multi-resolution graph structures, and approximate dynamic adjacency computation based on sparse top- k neighbor selection. Such strategies are complementary to AweHF and could be seamlessly integrated to improve scalability without altering the core modeling philosophy.

Cost of specialization

On relatively stationary datasets, AweHF maintains competitive predictive accuracy. Nevertheless, in such scenarios, its specialized modules, namely, AWD, TAN-DSPGC, and the bidirectional fusion mechanism, which are designed explicitly for non-stationary pattern modeling, may not be fully utilized. As a result, the associated computational cost and model complexity may outweigh their benefits in stationary conditions. This observation underscores the dual nature of AweHF. In particular, while it performs reliably on conventional traffic forecasting tasks, its architectural advantages are most pronounced in non-stationary environments that challenge existing models.

6. Conclusions

In this paper, we propose an AweHF to address the challenge of complex non-stationarity in MTS data. AweHF employs AWD to decouple sequences into high-frequency event and low-frequency trend components. This data-driven decomposition overcomes the limitations of fixed wavelet bases. To capture the distinct properties of the decomposed components, we apply TAN and DSPGC to model the spatiotemporal dynamics of high-frequency sequences. Meanwhile, a lightweight MLP is used to learn long-term dependencies of low-frequency components. Finally, a bidirectional interaction fusion mechanism integrates both components, leveraging their cross-scale characteristics to improve

overall prediction accuracy. Comprehensive experimental evaluations on six real-world datasets show that AweHF outperforms the current state-of-the-art baselines. While the dynamic graph design introduces quadratic computational complexity, our analysis shows a favorable performance–efficiency trade-off on practical-scale networks. Future work will explore scalable adaptive decomposition strategies and the integration of multi-source auxiliary information to further enhance robustness and applicability in complex real-world environments.

CRedit authorship contribution statement

Mengdi Gong: Writing – review & editing, Writing – original draft, Project administration, Methodology, Data curation, Conceptualization. **Chengci Wang:** Visualization, Validation, Methodology, Conceptualization. **Jie Yu:** Validation, Supervision. **Lingyu Xu:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

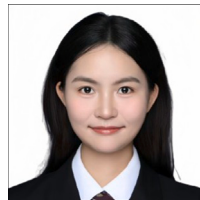
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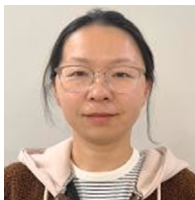
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